Redundancy-Reduction Improves World Model Sample Efficiency in Atari Games

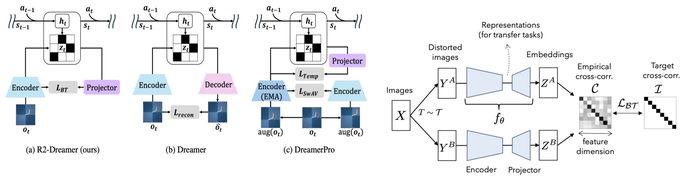
Kempner INSTITUTE
For the Study of Natural & Artificial Intelligence at Harvard University

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Background

- **World Models** pay a pixel tax. **DreamerV3** (Deepmind) prevents representation collapse (trivial solutions) by reconstructing pixels using a decoder, which is costly and sensitive to noise
- **Decoder-Free** variants exist. **R2-Dreamer** and **DreamerPro** replace reconstruction with other self-supervised techniques (**Barlow-Twins Redundancy Reduction** and **SwAV**, respectively).
- R2-Dreamer matches DreamerV3 on continuous control
- No comparison on **Discrete-Action, Low-Budget Atari** games



Methods

- **Benchmark:** 26 Atari Games, 100k agent steps only (~ 2 hours human Gameplay). Tests **Sample-Efficiency** in RL

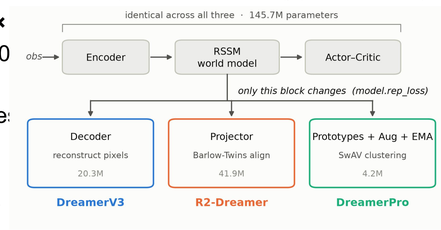


Calculate Human-Normalized Score: $HN = \frac{\text{score}_{\text{agent}} - \text{score}_{\text{random}}}{\text{score}_{\text{human}} - \text{score}_{\text{random}}}$

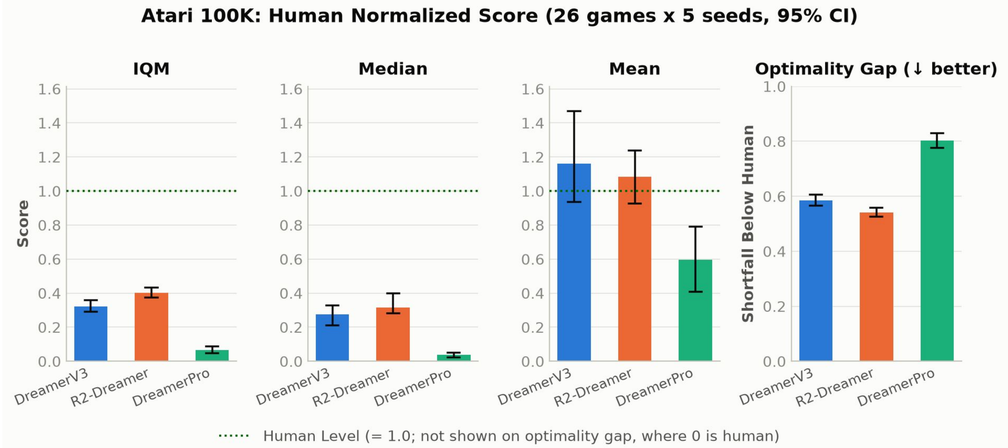
(0.0: Random Policy | 1.0 = Human)

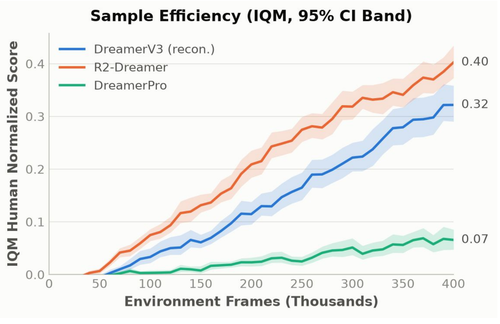
Setup:

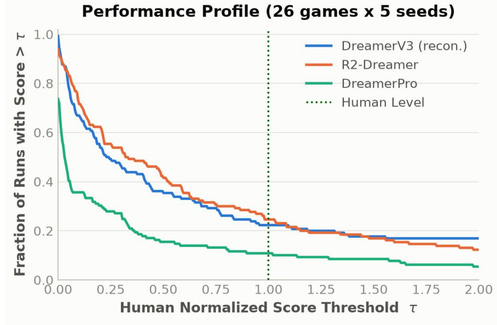
- **26 games × 5 seeds × models = 390 runs**, 40 frames each.
- 100 evaluation episodes with frozen policy
- 1 H100 GPU / run
- **41 Evaluations:** 1 per 10k steps

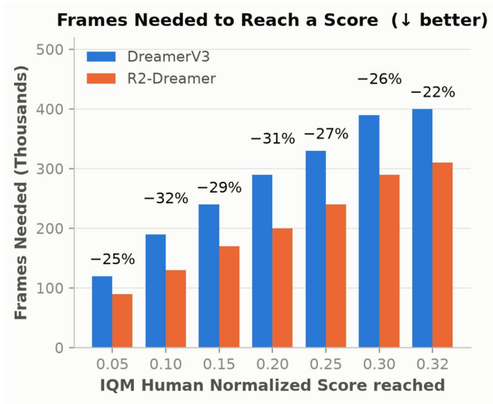


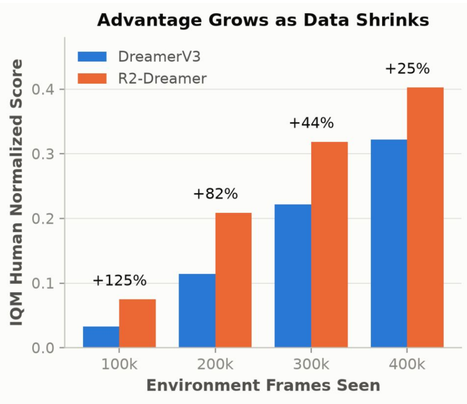
Results







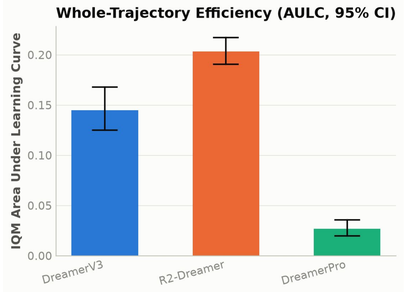




R2-Dreamer (Barlow-Twins instead of a Decoder) required on average 27% less frames to reach the same IQM Human Normalized Score on Atari100k, and the advantage was more when less data was available

Results

- R2-Dreamer achieves a **40% higher** AULC (Area Under Learning Curve), calculated based on 41 Scores captured every 10,000 frames
- Shows R2-Dreamer reached better scores earlier and maintained them



Conclusions & Limitations

- **Representation-space alignment beats pixel reconstruction under a tight budget.** IQM 0.402 vs 0.322, non-overlapping CIs; 40% higher AULC; DreamerV3's final score at 22.5% fewer frames.
- **The objective drived the result, not the missing decoder.** DreamerPro is equally decoder-free and lands at IQM 0.069 — nearly 6× lower, same backbone, protocol, and seeds.
- **The gain is mid-range, not frontier.** DreamerV3 still wins 14/26 games and the profiles cross at $\tau \approx 1.2$: R2-Dreamer learns the learnable middle faster and further.
- **Which ingredient is responsible is untested.** Decorrelation, standardization, and augmentation-free alignment remain confounded

Next Steps

- Run the models on **other benchmarks:** Crafter, DMLab, Robot Locomotion, etc.
- Further test the hypothesis by i**solating other differences** between R2Dreamer and DreamerPro
- Test the hypothesis on other World Models and RL agents by **replacing the decoders** with a Barlow-Twins (Redundancy Reduced) Self-Supervised Objective
- Run many other types of World Models (JEPA, Dreamer-variants) on Atari
- Test on **other architectures**



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References
[1] Morihira et al. *R2-Dreamer: Redundancy-Reduced World Models without Decoders or Augmentation.*
[2] Hafner et al. *Mastering Diverse Domains through World Models.* arXiv:2301.04104, 2024.
[3] Deng et al. *DreamerPro: Reconstruction-Free Model-Based RL with Prototypical Representations.*

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